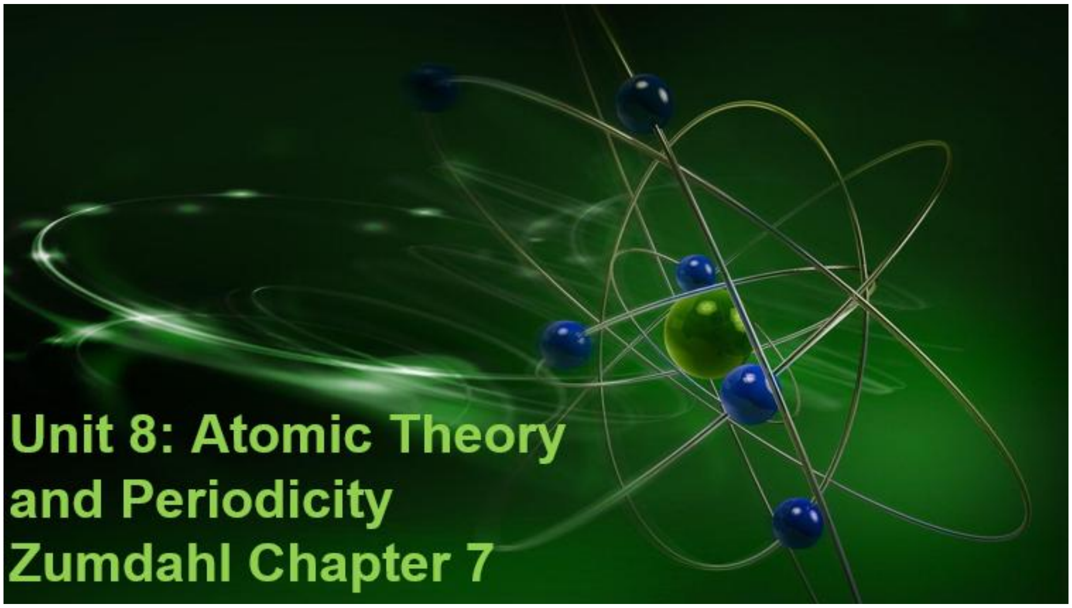
Name: ______________________________ Period: ____________



Unit 8: Atomic Theory and Periodicity

Unit Map

Textbook Reference: *Zumdahl*, Chemistry, Chapter 7

Date	Day	Learning Targets	Activities	Work on this
2/24-25	U7D9 U8D1	What is the structure of an atom?	Mass Spectrometry	
2/27-28	U7.2 Exam	What have you learned about buffers and titrations?		
3/2-3	U8D2	What is the electronic structure of an atom? What is photoelectron spectroscopy?	Nature of Light Electron Configuration PES- Photoelectron Spectroscopy	U8 Textbook Problem Set
3/6 C Day	U8D3	How can periodic trends be predicted?	Periodic Table Periodic Trends	
3/9-10	U8D4	How can periodic trends be predicted?		
3/11	SAT DAY	Juniors take SAT		
3/12-13	U8D5 U9D1	What have you learned about atomic theory? How do atoms bond?	Unit 8 QUIZ Bonding Lectures Begin	Finish assigned Bonding lectures for quiz after Spring Break!

Unit 8: Atomic Theory and Structure Learning Targets (CED Topic 1, partial 3)

1.2	**Mass Spectra of Elements**
1.2.A:	Explain the quantitative relationship between the mass spectrum of an element and the masses of the element's isotopes.
	1.2.A.1: The mass spectrum of a sample containing a single element can be used to determine the identity of the isotopes of that element and the relative abundance of each isotope in nature.
	1.2.A.2: The average atomic mass of an element can be estimated from the weighted average of the isotopic masses using the mass of each isotope and its relative abundance.
1.5	**Atomic Structure and Electron Configuration**
1.5.A:	Represent the ground-state electron configuration of an atom of an element or its ions using the Aufbau principle.
	1.5.A.1: The atom is composed of negatively charged electrons and a positively charged nucleus that is made of protons and neutrons.
	1.5.A.2: Coulomb's law is used to calculate the force between two charged particles.
	EQN: Fcoulombic $\propto$ q1q2 / r2
	1.5.A.3: In atoms and ions, the electrons can be thought of as being in "shells (energy levels)" and "subshells (sublevels)," as described by the ground-state electron configuration. Inner electrons are called core electrons, and outer electrons are called valence electrons. The electron configuration is explained by quantum mechanics, as delineated in the Aufbau principle and exemplified in the periodic table of the elements.
	1.5.A.4: The relative energy required to remove an electron from different subshells of an atom or ion or from the same subshell in different atoms or ions (ionization energy) can be estimated through a qualitative application of Coulomb's law. This energy is related to the distance from the nucleus and the effective (shield) charge of the nucleus.
1.6	**Photoelectron Spectroscopy**
1.6A	Explain the relationship between the photoelectron spectrum of an atom or ion and: i. The ground-state electron configuration of the species. ii. The interactions between the electrons and the nucleus.
	1.6.A.1: The energies of the electrons in a given shell can be measured experimentally with photoelectron spectroscopy (PES). The position of each peak in the PES spectrum is related to the energy required to remove an electron from the corresponding subshell, and the relative height of each peak is (ideally) proportional to the number of electrons in that subshell.
1.7	**Periodic Trends**
1.7A	Explain the relationship between trends in atomic properties of elements and electronic structure and periodicity.
	1.7.A.1: The organization of the periodic table is based on patterns of recurring properties of the elements, which are explained by patterns of ground-state electron configurations and the presence of completely or partially filled shells (and subshells) of electrons in atoms. ★
	1.7.A.2: Trends in atomic properties within the periodic table (periodicity) can be predicted by the position of the element on the periodic table and qualitatively understood using Coulomb's law, the shell model, and the concepts of shielding and effective nuclear charge. These properties include:
	i. Ionization energy ii. Atomic and ionic radii iii. Electron affinity iv. Electronegativity.
	1.7.A.3: The periodicity (in 1.7.A.2) is useful to predict / estimate values of properties in the absence of data.
3.12	**Properties of Photons**
3.12A	Explain the properties of an absorbed or emitted photon in relationship to an electronic transition in an atom or molecule.
	3.12.A.1: When a photon is absorbed (or emitted) by an atom or molecule, the energy of the species is increased (or decreased) by an amount equal to the energy of the photon.
	3.12.A.2: The wavelength of the electromagnetic wave is related to its frequency and the speed of light by the equation:
	EQN: $c = \lambda v$.
	The energy of a photon is related to the frequency of the electromagnetic wave through Planck's equation: **EQN: $E = hv$.**

ATOMIC STRUCTURE REVIEW

Parts of Atoms:

Most people already know that the atom is made up of three main parts, the _______________ and ______________ in the **nucleus** and the ______________ somewhere outside of the **nucleus**.

Let's summarize:

	proton	neutron	electron
symbol			
charge			
location			
mass			
size (see below)	10^{-15} m	10^{-15} m	10^{-18} m

The atom is often represented as a miniature solar system. Draw an atom of beryllium-9 below showing all the correct particles:

The **mass** of the atom is due to the

____

The **size** of the atom is due to the

____

How Many Particles in Each Atom?

The particle that defines the identity of an atom is the _____________. (shown on the periodic table)

Every hydrogen atom has ___ proton.

Every magnesium atom has ___ protons.

Any atom that has 23 protons is _________________.

Any atom that has 92 protons is _________________.

The mass of an atom is mostly from the ___________ and ____________.

Find O on the periodic table. Its mass is ______ amu.

It has ___ protons. It must have ___ neutrons.

Electrically neutral atoms (as opposed to ions) have one electron for every proton.

Fill in this chart for these neutral atoms. Use the nearest whole number for mass:

Atom	Mass	protons	neutrons	electrons
He				
Si				
Be				
H				
Rn				
Ar				
F				
Pb				

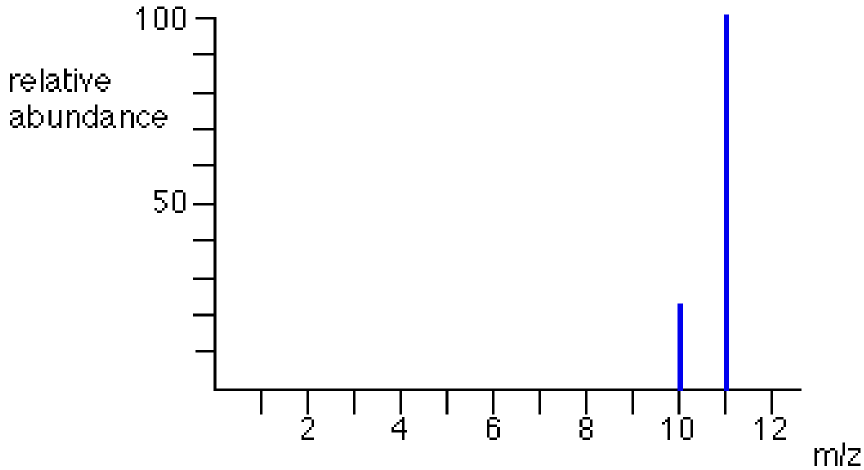
If the mass is not close to a whole number, it is because the atom has several _____________. These are atoms with the same number of ___________ but different numbers of _____________.

Chlorine has two isotopes: Cl-35 (___ p+ & ___ n°) and Cl-37 (___ p+ & ___ n°).

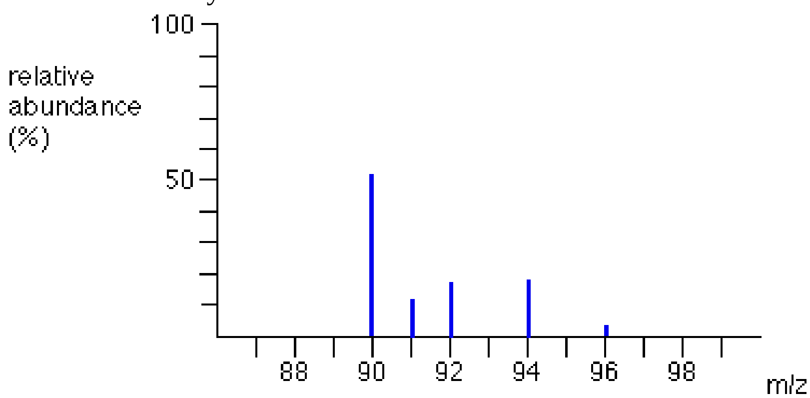
Practice with Average Atomic Mass Spectrum

1. Silver has two isotopes, Ag-107 (relative isotopic mass is 106.91) and Ag-109 (relative isotopic mass is 108.90). Ag-107 is 51.84% abundant in nature while Ag-109 is 48.16%. Calculate the average atomic mass of silver.

2. Below is the mass spec data for an element. Determine the percent abundance and the average atomic mass to this data. Identify the element.



3. Below is the mass spec data for an element. Determine the percent abundance and the average atomic mass to this data. Identify the element.



The Electromagnetic Spectrum

The electromagnetic spectrum represents the range of ________________________ __________________.

It can be measured with three variables: ____________________ (____), ____________________ (____), ____________________ (____).

$c = \lambda v$

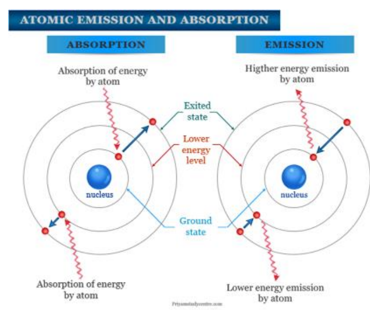
- The frequency (v) of a wave is the number of waves to cross a point in 1 second (units are Hertz – cycles/sec or sec^{-1})
- λ is the wavelength- the distance from crest to crest on a wave
- $c = 3.00 \times 10^8$ m/s, speed of light
- $c = 2.998 \times 10^8$ m/s, speed of light?

$E = hv$

- v is still the frequency (Hertz or s^{-1})
- E is the energy of the photon (Joules)
- $h = 6.626 \times 10^{-34}$ Js, Planck's constant

Photoelectric effect- shine light on a metal and electrons can be ejected from it

Atoms and Light:



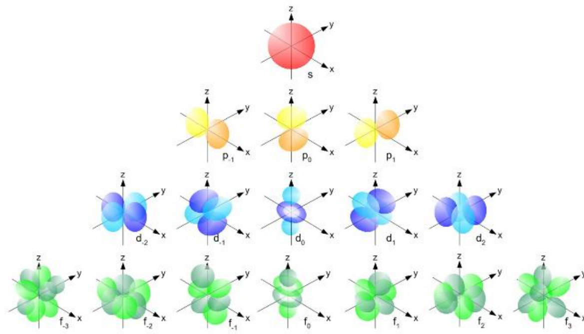
- Electrons in atoms can ___________ light energy- sometimes enough to free them! (Photoelectric effect!)
- Sometimes it is enough to take an electron from its _________________ __________________ (least energetic) to an __________________ __________________ (energized!)
- Each element absorbs only certain frequencies of light, called the ______________________ ___________________.
- Once the electrons are in the excited state they can '______________________' back to the ground state be _________________________ light
- This is called the '____________________________ ______________________' and it is different for each element. Emission spectra can therefore can be used to ____________________ types of element.
- The light ___________________ *or* ___________________ has an energy, frequency, and wavelength found with: $c = \lambda v$ and $E=hv$

Practice:

Calculate the wavelength and energy of a photon that moves from the 1s orbital to the $2p^1$ orbital in an atom with a frequency of 2.9×10^{15} Hz. How much energy would a mole of these photons have in kJ/mol?

Those exciting electrons!

- We just learned that electrons could be excited from their ____ to a higher level ____by the absorption of light.
- The excited electrons can then release ______________________ (in the form of light) and return to their ground state.
- The ______________________of energy absorbed or released are very ______________________because the energies of electrons in ______________________are very predictable.
- Electrons orbit around the nucleus in nonrandom patterns called '______________________.
- Each atom has many orbits of differing energies. The ______________________ the energy, the ______________________ to the nucleus. Each atom has a ______________________ set of energies.



- There are rules for filling orbitals with electrons.
- Orbitals are NOT ____________________________ or _____________________________________.
- They are spaces- _______________________________ -of probability where the atom is most likely found.

Electron Configurations

- The ways in which electrons are ____________________________ in various orbitals around the nuclei of atoms are called electron configurations.
- Three rules—the Aufbau principle, the Pauli exclusion principle, and Hund's rule—tell you how to find the electron configurations of atoms.
- **Aufbau Principle-** According to the aufbau principle, electrons occupy the orbitals of lowest energy first
- **Pauli Exclusion Principle-** According to the Pauli exclusion principle, an atomic orbital may describe at most two electrons. To occupy the same orbital, two electrons must have opposite spins (two seats in a loveseat!)
- **Hund's Rule-** Hund's rule states that electrons occupy orbitals of the same energy in a way that makes the number of electrons with the same spin direction as large as possible. (No unnecessary loveseat sharing!)
- The Periodic Table is arranged according to electron configuration… read it like a book
- Nobel Gas Configuration- Lists core electrons together with the use of the highest filled nobel gas, then lists the valence electrons

Practice:

- Na
- C
- Xe
- S^{-2}

- Orbital diagrams show the electrons as half arrows in a representation of their locations.
- If the electron configuration is like the ROSTER (list) of electrons in the atom, then the orbital diagram is like a SEATING CHART
- Rules- fill lowest levels first, NO SHARING unless they have to, all same spin until sharing is forced

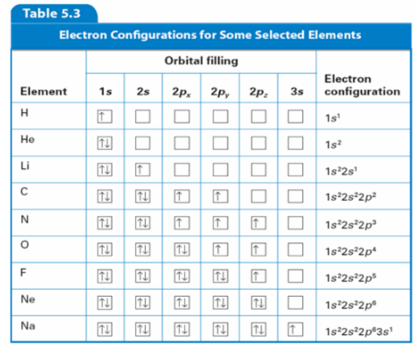


Table 5.3

Electron Configurations for Some Selected Elements

Element	Orbital filling: 1s	2s	$2p_x$	$2p_y$	$2p_z$	3s	Electron configuration
H	↑						$1s^1$
He	↑↓						$1s^2$
Li	↑↓	↑					$1s^2 2s^1$
C	↑↓	↑↓	↑	↑			$1s^2 2s^2 2p^2$
N	↑↓	↑↓	↑	↑	↑		$1s^2 2s^2 2p^3$
O	↑↓	↑↓	↑↓	↑	↑		$1s^2 2s^2 2p^4$
F	↑↓	↑↓	↑↓	↑↓	↑		$1s^2 2s^2 2p^5$
Ne	↑↓	↑↓	↑↓	↑↓	↑↓		$1s^2 2s^2 2p^6$
Na	↑↓	↑↓	↑↓	↑↓	↑↓	↑	$1s^2 2s^2 2p^6 3s^1$

Light, energy, wavelength, frequency, and atoms

Formulas and Constants				
$c = \lambda\upsilon$	$\upsilon = \frac{c}{\lambda}$	$\lambda = \frac{c}{\upsilon}$	$E = h\upsilon$	$E = \frac{hc}{\lambda}$
$c = 2.998 \times 10^8$ m/s	$h = 6.626 \times 10^{-34}$ J·s			
1 nanometer = 1×10^{-9} m	1 meter = 1×10^{9} nm			

1. Perform the following conversions:

   519 nm = ____________________m 2.12 x 10^{-9} m = ____________________ nm

   4 x 10^{-6} m = __________________nm 46700 nm = ______________________ m

2. List all electromagnetic radiations from low energy to high. (hint: Red Martians Invade Venus Using Xray Guns)

3. Violet light has a wavelength of about 410 nm. What is its frequency? Calculate the energy of one photon of violet light. What is the energy of 1.0 mol of violet photons?

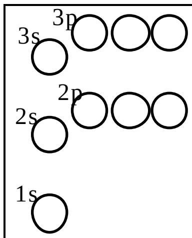
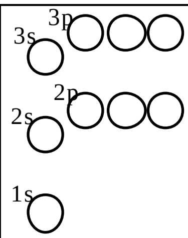
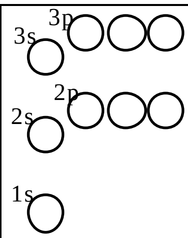
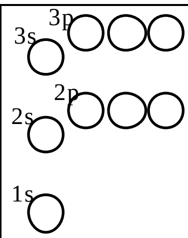
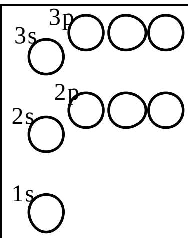
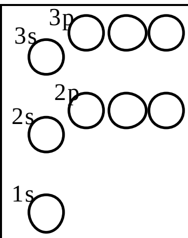
4. The most prominent line in the spectrum of neon is found at 865.438 nm. Other lines are found at 837.761 nm, 878.062 nm, 878.438 nm, and 1885.387 nm.

(a) Which of these lines represents the most energetic light?

(b) What is the frequency of the most prominent line? What is the energy of one photon of this wavelength?

<u>WRITING ELECTRON CONFIGURATIONS</u>

For each given element, fill in the orbital diagram and then write the electron configuration for the element.

1.	2.	3.	4.	5.	6.
					
Element: Ar # of e⁻'s: ___	Element: Mg # of e⁻'s: ___	Element: N # of e⁻'s: ___	Element: Li # of e⁻'s: ___	Element: P # of e⁻'s: ___	Element: Cl # of e⁻'s: ___

Write the electron configurations of each of these in **long form** and **short form**:

1. Ar

   Ar

2. Mg

   Mg

3. N

   N

4. Li

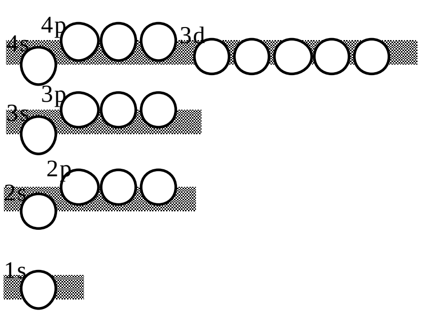
   Li

5. P

   P

6. Cl

   Cl

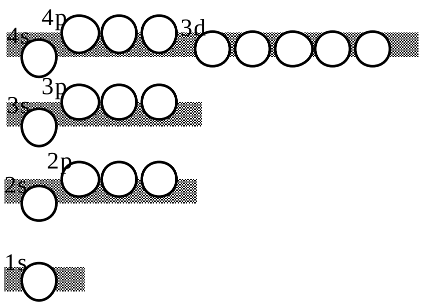


7. Fill in the orbital diagram for the element, Fe, and write the electron configuration of Fe in the long and short form.

   Fe

   Fe

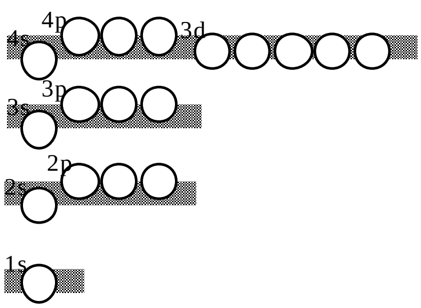
A few elements do not follow the “rules”. There is some lowering of the energy of the atom by completely filling or half-filling the five d-orbitals.



8. Fill in the orbital diagram for the element, Cu, and write the electron configuration of Cu in the long and short form.

Cu

Cu



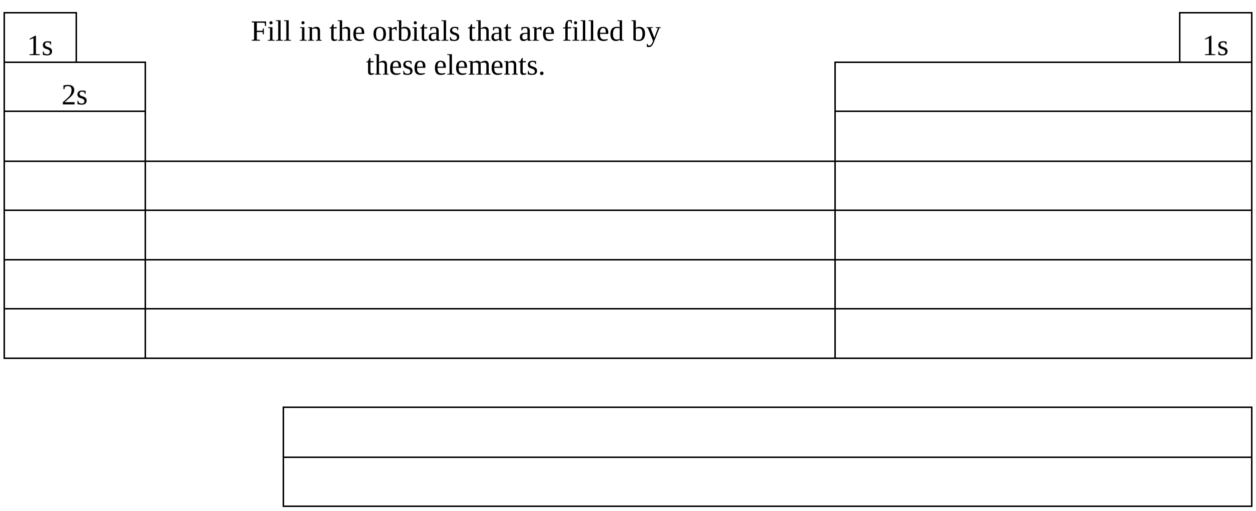
9. Fill in the orbital diagram for the element, Cr, and write the electron configuration of Cr in the long and short form.

Cr

Cr

Shade in the six elements that do not follow the Aufbau Principle:

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg



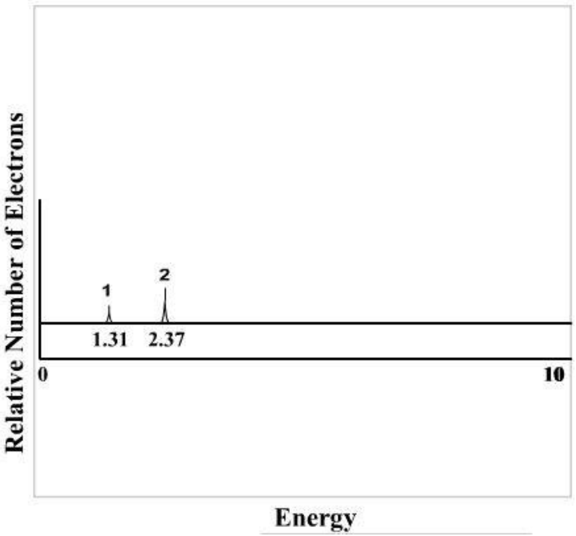
10. Write the orbital occupied by the last electron of each of the following elements:

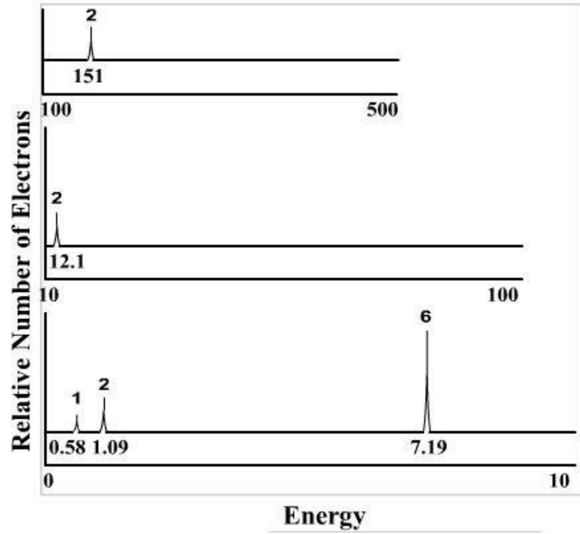
As	W	Li	U	O	Rn	V

Photoelectron Spectroscopy

Refer to the PES spectra below. For the spectrum on the left, there are two elements overlayed: the integration peak with 1 electron is that of hydrogen, whereas the other peak reflecting two electrons represents helium.

The spectrum on the right is used for questions 3-5.





For the left hand spectrum:

1) How does this diagram show that the strength of attraction between protons and electrons as their amounts increase?

2) What does this diagram suggest about the energy needed to remove an electron from a half-filled orbital versus a filled orbital?

For the right hand spectrum:

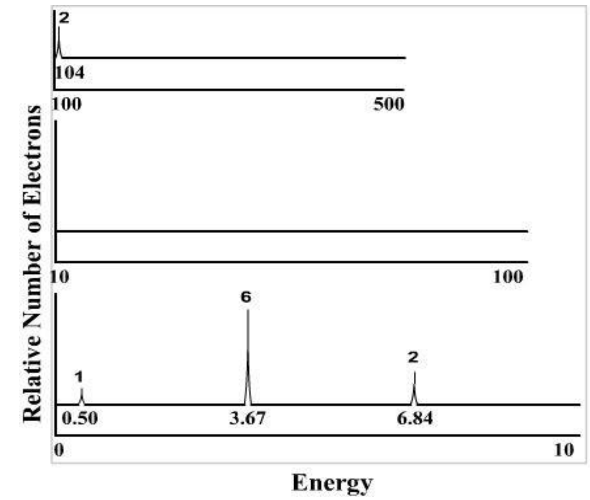
3) Which energy value is associated with the 1s electrons? ___________________

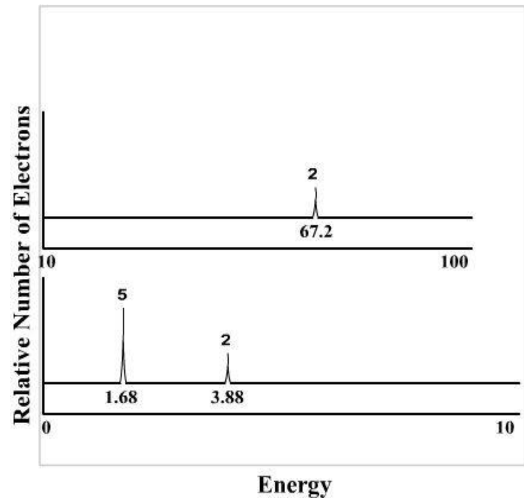
4) What element is represented by the right hand spectrum? __________________

5) Explain the following questions:
   a) Why does it take a significantly greater amount of energy to remove core electrons as compared to valence electrons?

   b) Why do the integration peaks have different heights?

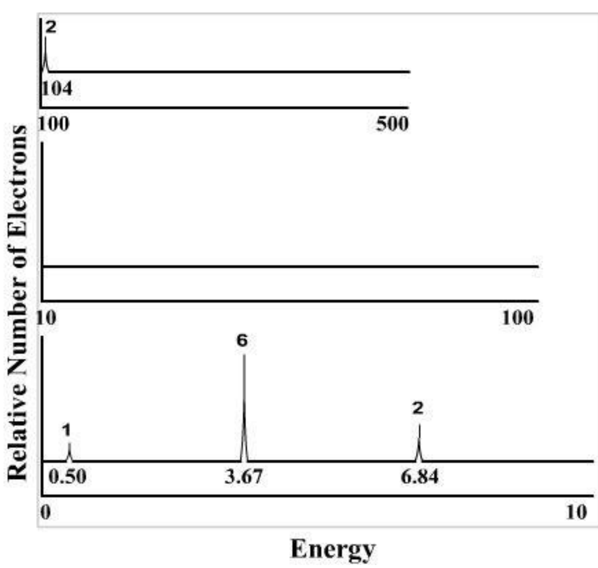
Refer to the PES spectra below. Identify the elements for each and then proceed to the prompts.

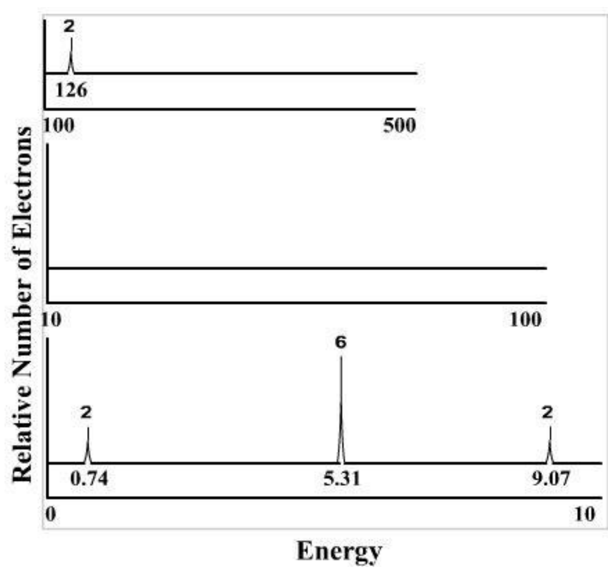




6) Based on the spectra above, discuss why the element in the left spectrum is better suited for forming a positive ion (cation), whereas the element in the right spectrum is better suited for forming a negative ion (anion).

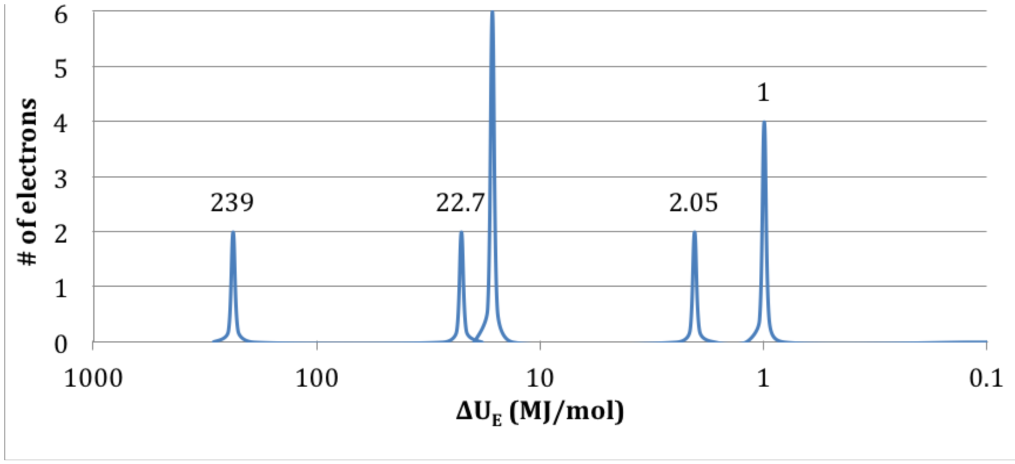
Refer to the PES spectra below. Identify the elements for each and then proceed to the prompts.





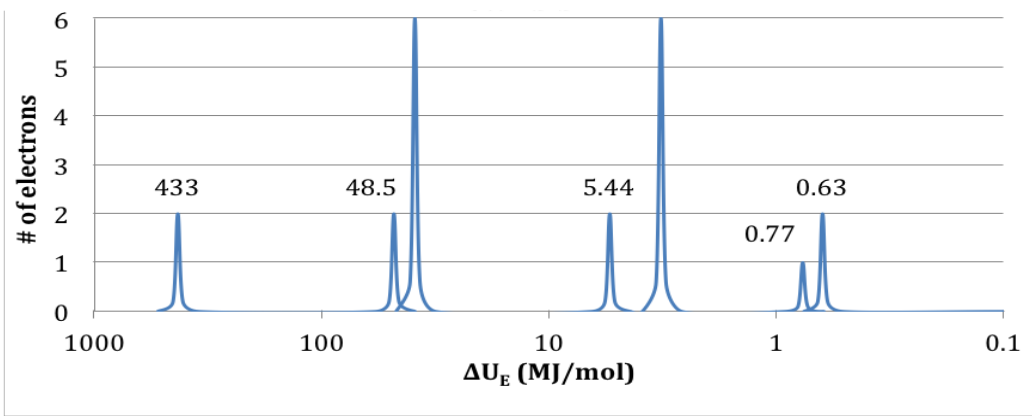
7) Based on the spectra above, which element is more likely to form a cation with a +2 charge? Why?

8) Identify the following elements based on their PES data:



a)

The element is:________________________________



b)

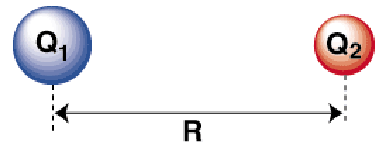
The element is :___________________________________

Periodic Trends

Periodic Trends are specific ______________ that are present in the periodic table that illustrate different aspects of a certain element.

Almost all of the properties that are asked about in exam questions rely on Coulombic (electrostatic) attraction between outer electrons and the nucleus. Hence, you should refer to Coulomb's law in your answers!

$$E \propto \frac{Q_1 Q_2}{r}$$



E = ionization energy, the energy needed to _____________ the outermost electron.
Q_1 = charge of an electron, -1.
Q_2 = effective nuclear charge (________) of protons in nucleus.
r = distance between charged particles, which can be approximated by the period (n, energy level).

In short, the **energy of attraction or repulsion** between charged particles is:

- ___________________ proportional to the **magnitude** (size) of the charges
- ___________________ proportional to the **distance** between the charges

All Periodic Trends can be understood in terms of Coulomb's Law(!)

1. Electrons are attracted to the protons in the nucleus of an atom.
   a. The ______________ an electron is to the nucleus, the **MORE** strongly it is attracted.
      i. Distance between electrons and the nucleus can be approximated by _____ (main energy level)
   b. The __________ protons in a nucleus, the **MORE** strongly an electron is attracted.
2. Electrons are repelled by other electrons in an atom.
   a. If other electrons are _______________ a valence electron and the nucleus, the valence electron will be **LESS** strongly attracted to the nucleus (this is known as _________________).

Effective Nuclear Charge (________)**:** net positive charge experienced by electrons

→ Attraction of a valence electron to a proton is partially shielded by the inner shell (__________) electrons, and so the farther away an electron is from the nucleus, the less ______ charge it feels from the nucleus!

→ Thus, following Coulomb's Law:
- The nucleus of atoms with a **higher Z_{eff}** (at the same energy level, n) will be ______________ attractive to their valence electrons
- The nucleus of atoms with a **lower Z_{eff}** (at the same energy level, n) will be ______________ attractive to their valence electrons

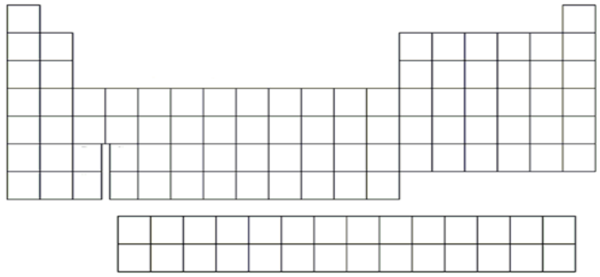
→ At the ________ energy level, Z_{eff} is directly proportional to the number of _________!

NOTE: Energy level trumps Z_{eff}!

If you're comparing elements that have different n AND different Z_{eff}, _____ matters more!!

Trendy Trends: Periodic Trends to Know

<u>Atomic radius</u> (size of atom): distance between the ____________ and valence electrons.



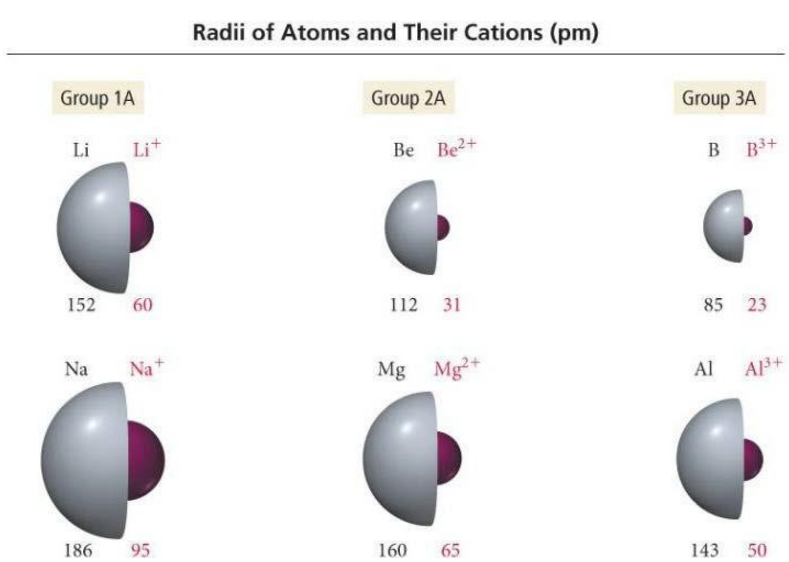
<u>Ionic radius</u>: distance from the nucleus to valence electrons in a charged _______.

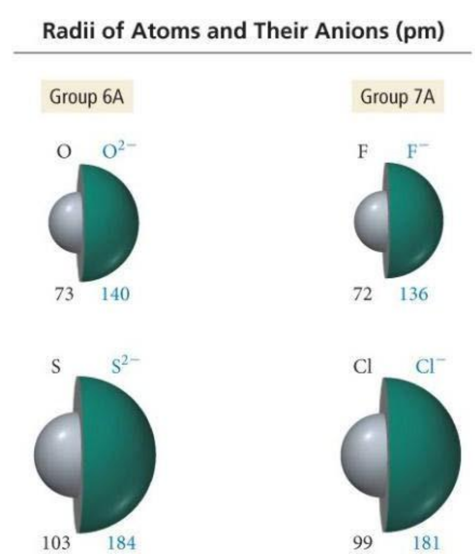
Most comparisons between <u>an atom and their ion</u> or <u>ions of the same atom</u> can be explained by **e^-/e^- repulsion**:

- Positively charged cations are **SMALLER** than the neutral atom because _______________ electrons in the outermost shell results in _________ e^-/e^- repulsion, thus valence electrons are ____________ to the nucleus.
- Negatively charged anions are **LARGER** than the neutral atom because __________ electrons in the outermost shell results in _________ e^-/e^- repulsion, thus valence electrons are ______________ from the nucleus.

Only one type of comparison between an atom and their ion(s) should **NOT** utilize the e^-/e^- repulsion argument:

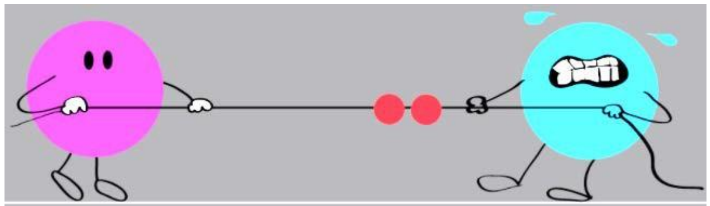
- Metals cations which have lost sufficient electrons such that their valence electrons are now in a _____________ energy level (n). Examples: Sr vs Sr^{2+}, Na vs Na^+, Al^{2+} vs Al^{3+}, etc.
- Why? Because **n matters more than Z_{eff}!!!** If a species has their outermost electrons on a lower energy level (n), their valence electrons are ____________ to and thus _________ attracted to the nucleus.





Electronegativity: attraction of an atom for pair of ____________ level electrons in a covalent bond with another atom.

Think of the atoms as playing "tug of war" with their _____________ shell electrons!



Electronegativity of an atom is determined by combining the following two trends:

Ionization Energy vs Electron Affinity

Ionization Energy	**Electron Affinity**
Energy required to remove an electron	Energy change when electron is added to an atom
$X(g) + energy \rightarrow X^{+}(g) + e^{-}$	$X(g) + e^{-} \rightarrow X^{-}(g) + energy$
endothermic (+kJ/mol)	first electron added is always exothermic (−kJ/mol)

Higher attraction between nucleus and electron = ______ ionization energy and _____ electron affinity!

Metallic vs Non-Metallic Character

Metallic Character	**Non-Metallic Character**
Metals react by losing electrons	Non-metals react by gaining electrons
How easy it is to remove an electron from an atom or ion	How hard is it to remove an electron from an atom or ion
___ Ionization Energy = ___ Metallic Character	___ Ionization Energy = ___ Non-Metallic Character

Let's Practice!

O Oxygen	F Fluorine
S Sulfur	Cl Chlorine

Consider the elements shown above. Which element has the:

a) Highest ionization energy?__________________

b) Highest electron affinity?___________________

c) Highest Metallic Character?_________________

d) Highest Non-Metallic Character?_____________

Successive Ionization Energies

Removing one electron, and then another electron, and then another electron…

- 1st Ionization Energy: (________) energy required to remove the first (highest energy) electron
- 2nd Ionization Energy: (________) energy required to remove the second electron (second highest energy)
- Each additional electron requires **MORE** energy to remove than the previous one:

$$IE_1 < IE_2 < IE_3 \text{ (etc.)}$$

- Valence (outer) electrons require much **LESS** energy to remove than core (inner) electrons

$$IE_{valence} << IE_{core}$$

TABLE 8.1 Successive Values of Ionization Energies for the Elements Sodium through Argon (kJ/mol)

Element	IE_1	IE_2	IE_3	IE_4	IE_5	IE_6	IE_7
Na	496	4560					
Mg	738	1450	7730		Core electrons		
Al	578	1820	2750	11,600			
Si	786	1580	3230	4360	16,100		
P	1012	1900	2910	4960	6270	22,200	
S	1000	2250	3360	4560	7010	8500	27,100
Cl	1251	2300	3820	5160	6540	9460	11,000
Ar	1521	2670	3930	5770	7240	8780	12,000

You can identify an element from the pattern of successive ionization energies!

FR Practice

IE_1	IE_2	IE_3	IE_4	IE_5
801 kJ/mol	2,426 kJ/mol	3,660 kJ/mol	24,682 kJ/mol	32,508 kJ/mol

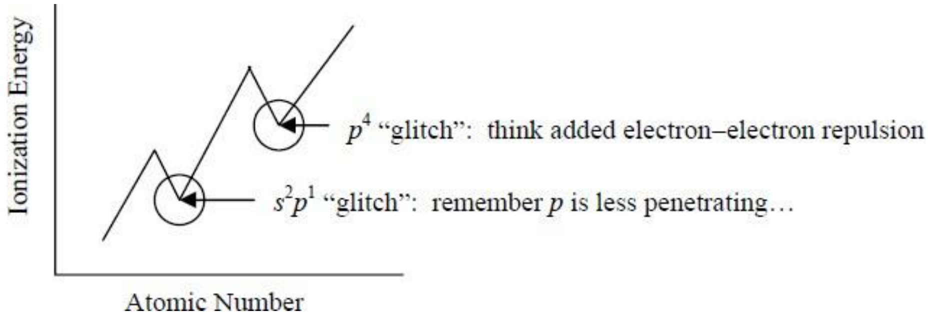
1. Which element from Period 2 does the table of successive ionizations energies above represent? Explain.

2. Would it require more energy to remove an electron from an F^+ ion or an F^{2+} ion? Justify your answer using Coulomb's Law.

<u>Ionization Energy Exceptions</u>:

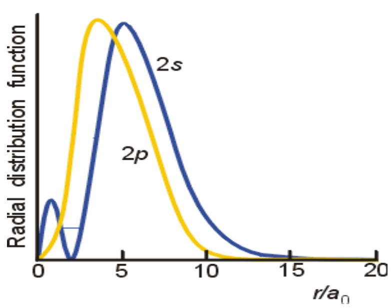
You MUST know these!

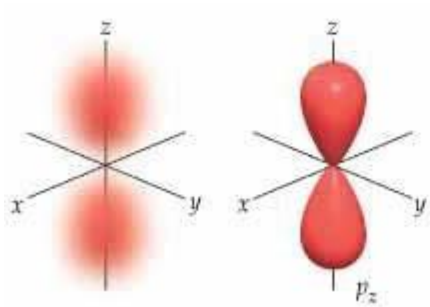
*these exceptions only matter when both elements are found on the same <u>period</u> (energy level).

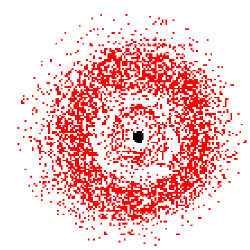


1. **<u>Group 5 vs Group 6</u>: "p^4 glitch"** (Example: oxygen vs nitrogen): It's easier to remove an electron from oxygen ($2p^4$) because of the _____________ of the paired electrons (whereas the $2p^3$ electrons in nitrogen (Group 5) are all unpaired).

2. **<u>Group 2 vs Group 3</u>: "s^{2}p^1 glitch"** (Example: beryllium vs boron): It's easier to remove an electron from boron ($2s^22p^1$) because it's being removed from the p orbital, and p orbitals do not ___________________ the nuclear region as greatly as the s orbitals, so electrons in a p orbital are not as tightly held.





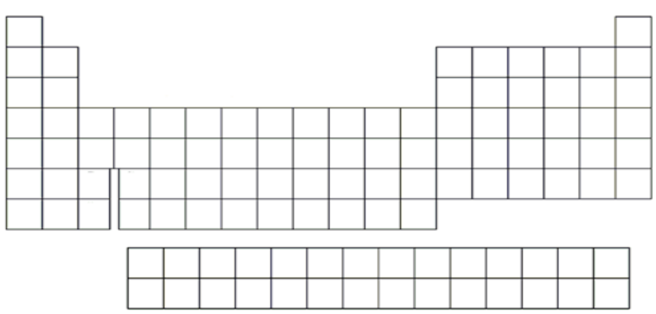


a 2s orbital

s orbital: higher nuclear penetration D more electron density in greater proximity to nucleus

BE CAREFUL! This is NOT about distance between electron and nucleus - on average, electrons in a 2s orbital and 2p orbital are the same distance from the nucleus.

Ionization Energy Trend

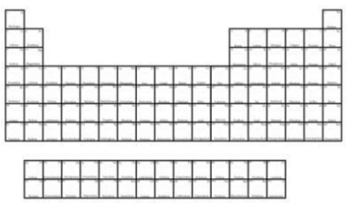
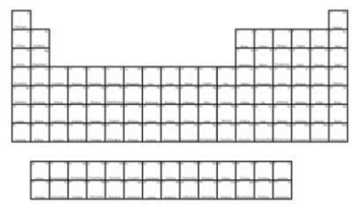
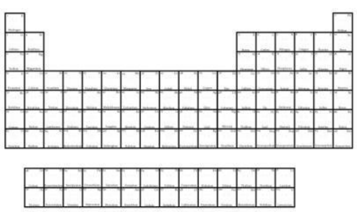
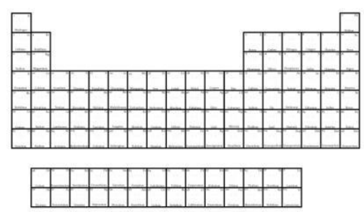
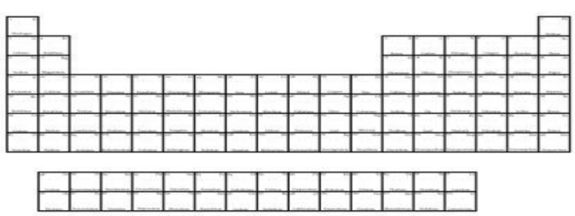


One last trend:

- **<u>Reactivity</u>** depends on whether the element reacts by losing electrons (metals) or gaining electrons (nonmetals).
    - <u>Metals are MORE reactive as you move down a column</u>: because metals ___________ electrons as they react, **LESS** attraction between valence electrons and the nucleus will result in a ___________ reactive metal.
    - <u>Non-metals are LESS reactive as you move down a column</u>: because non-metals ___________ electrons as they react, **LESS** attraction between valence electrons and the nucleus will result in a ___________ reactive non-metal.

In Summary

Draw arrows in the direction that each trend increases!

Atomic Radius Trend 	Metallic Character Trend 
Electronegativity Trend 	Ionization Energy Trend 
Reactivity Trend 	

Multiple Choice Practice

	Ionization Energy
First	740 kJ/mol
Second	1,450 kJ/mol
Third	7,730 kJ/mol
Fourth	10,540 kJ/mol
Fifth	13,610 kJ/mol

1. Given the table of ionization energies for the unknown element M shown above, which of the following is the most probable empirical formula for the compound element M would form with fluorine?

   a. MF b. MF_2 c. MF_3 d. MF_4

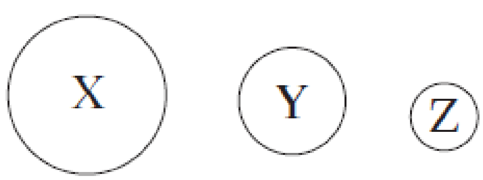
2. Why does an ion of phosphorus, P^{3-}, have a larger radius than a neutral atom of phosphorus?

   a. There is a greater Coulombic attraction between the nucleus and the electrons in P^{3-}.
   b. The core electrons in P^{3-} exert a weaker shielding force than those of a neutral atom.
   c. The nuclear charge is weaker in P^{3-} than it is in P.
   d. The electrons in P^{3-} have a greater Coulombic repulsion than those in the neutral atom.

3. Which of the following, sodium or magnesium, has greater metallic character and why?

   a. Sodium, because sodium has fewer principal energy levels than magnesium.
   b. Sodium, because sodium has a lower first ionization energy than magnesium.
   c. Magnesium, because of the repulsion of magnesium's paired 4s electrons.
   d. Magnesium, because magnesium has a greater effective nuclear charge than sodium.

4. The diagram below shows the relative atomic sizes of three different elements from the same period. Which of the following statements must be true if trend exceptions are ignored?



   a. The effective nuclear charge will be the greatest in element X.
   b. The first ionization energy will be greatest in element X.
   c. The electron shielding effect will be greatest in element Z.
   d. The electronegativity value will be greatest in element Z.

5. Which of the following isoelectric species has the smallest radius?

   a. S^{2-} b. Cl^- c. Ar d. K^+

6. Nitrogen's electronegativity value is between those of phosphorus and oxygen. Which of the following correctly describes the relationship between the three values?

   a. The value for nitrogen is less than that of phosphorus because nitrogen is larger, but greater than that of oxygen because nitrogen has a greater effective nuclear charge.
   b. The value for nitrogen is less than that of phosphorus because nitrogen has fewer protons, but greater than that of oxygen because nitrogen has fewer valence electrons.
   c. The value for nitrogen is greater than that of phosphorus because nitrogen has fewer electrons, but less than that of oxygen because nitrogen is smaller.
   d. The value for nitrogen is greater than that of phosphorus because nitrogen is smaller, but less than that of oxygen because nitrogen has a smaller effective nuclear charge.

7. Which of the following, argon or krypton, has a higher IE_1 and why?

   a. Argon, because argon has fewer principal energy levels than krypton.
   b. Argon, because argon has a larger effective nuclear charge than krypton.
   c. Krypton, because krypton has more principal energy levels than argon.
   d. Krypton, because krypton has a larger effective nuclear charge than argon.

8. The first five ionization energies for an element are listed in the table below.

First	Second	Third	Fourth	Fifth
8 eV	15 eV	80 eV	109 eV	141 eV

   Based on the ionization energy table, the element is most likely to be

   a. Sodium b. Magnesium c. Aluminum d. Silicon

9. Which of the following statements is true regarding sodium and chlorine?

   a. Sodium has a greater electronegativity and a larger first ionization energy.
   b. Sodium has a larger first ionization energy and a larger atomic radius.
   c. Chlorine has a larger atomic radius and a greater electronegativity.
   d. Chlorine has a greater electronegativity and a larger first ionization energy.

10. Consider the halogens chlorine and bromine. Which has a larger atomic radius and why?

   a. Chlorine has a larger atomic radius because it has an increased number of principal energy levels.
   b. Chlorine has a larger atomic radius because it has a higher effective nuclear charge.
   c. Bromine has a larger atomic radius because it has an increased number of principal energy levels.
   d. Bromine has a larger atomic radius because it has a higher effective nuclear charge.

11. Which of the following species is NOT isoelectronic to Br^-?

   a. Se^{2-} b. Kr c. Rb^+ d. K^+

How to Answer Periodic Trends Free Response Questions

Justifying all of the trends on the periodic table can be simplified using these two generalizations:

1. Use ***number of protons*** *(or Z_{eff})* to justify trends **<u>across a period</u>**.
2. Use **<u>increased distance (greater value of *n*)</u>** to justify trends **<u>down a group</u>**.

How to Earn Full Points on Periodic Trends Problems

Follow these three steps EVERY time you answer a periodicity question!

1) Locate *both* elements on the periodic table and **<u>state</u>** the principal energy level (*n*) and the sublevel containing the valence electrons for *each* element.
2) Do they have the same or different ____ values?
3) If same *n*: argue with number of ____________; if different *n*: argue with *n* vs. *n* ______________.

REMEMBER: a trend is not an explanation!

Simply identifying a trend (atomic radius decreases as you move from left to right across a period, electronegativity decreases as you move down a column, etc) earns _______ points!

Avoid Losing Easy Points

1. When explaining, you <u>must</u> refer to **<u>ALL</u>** species (atoms, ions) referenced in the question, or you will not get full credit.
2. Read the question: justify with "principles of atomic structure" or "Coulomb's Law" (it will always be one or the other 😊 😊).

Specific Question Types

1. Comparisons between ___________________ species: explain with **number of p^+**
   a. Isoelectronic species with ________ protons are **SMALLER** because the valence electrons are ________ attracted to and thus **CLOSER** to the nucleus.
   b. Isoelectronic species with ________ protons are **LARGER** because the valence electrons are ________ attracted to and thus **FARTHER** from the nucleus.
2. Comparisons between an atom and its ion/ions of the same atom, ___________ n: explain with **e^-/e^- repulsion**
   a. Positively charged cations are **SMALLER** than the neutral atom because of_______ e^-/e^- repulsion, thus valence electrons are **CLOSER** to the nucleus.
   b. Negatively charged anions are **LARGER** than the neutral atom because of_______ e^-/e^- repulsion, thus valence electrons are **FARTHER** from the nucleus.
3. Comparisons between an atom and its ion/ions of the same atom, ________________ n: explain with **distance**
   a. If a species has their outermost electrons on a lower energy level (n), their valence electrons are _____________ to and thus _________ attracted to the nucleus.

Guided Practice: How will an answer be different when explained in terms of "atomic structure" vs "Coulomb's Law"?

1. Which ion would require more energy to remove an electron: Mn^{3+} or Mn^{4+}? Explain using principles of atomic structure.

2. Which ion would require more energy to remove an electron: Mn^{3+} or Mn^{4+}? Be sure to reference Coulomb's Law in your explanation.

Independent Practice

1. Using principles of atomic structure, explain why the Na^+ ion is larger than the Li^+ ion.

2. Which has a higher metallic character, an atom of lithium or an atom of boron? Explain your answer in terms of Coulomb's Law.

3. Which would be greater for the aluminum atom, its first ionization energy or its second ionization energy? Explain using principles of atomic structure.

4. Which element, Cl or Br, has a greater first ionization energy? Justify your answer using Coulomb's Law.